

Haas Doubler

Short-delay stereo widening and doubling, with detune and drift.



Overview

Haas Doubler widens a sound with short delays. A copy delayed by a few milliseconds and placed on the other side is heard as the same sound, but wider - the Haas effect. Add a little drift and detune and it becomes a convincing double-tracked part.

Three layouts are built in. L/R is the classic: the dry signal on one side, the delayed copy on the other. LCR keeps the dry signal in the centre with delayed copies hard left and right. SPREAD uses the delayed copies only, left and right. In LCR and SPREAD the right copy's delay is the left one's times a multiplier (x2 to x6), so the two sides never line up.

DRIFT makes the delay time, the pan position and the level of each copy wander slowly and independently, like a second take. DETUNE pitch-shifts each side by a few cents for the classic detuned-doubler width.

Reach for it for

- Widening mono vocals, guitars and synths

- Fake double-tracking on lead vocals
- A wet-only widener on an FX send (DRY off)
- Detuned stereo width on pads and leads
- Wobbly, tape-style double tracking

Controls

Modes (header)

L/R	The classic Haas: dry signal on one side, the delayed copy on the other.
LCR	Dry signal in the centre, delayed copies hard left and right (the right copy at DELAY times the multiplier).
SPREAD	Delayed copies only, left and right (the right at DELAY times the multiplier). No dry signal in the centre.

Display

Stereo stage	Shows where each copy sits between L and R, its delay time and (by the size of the dot) its level, moving as it drifts. DRY marks the dry signal in the centre in LCR, or on its side in L/R.
INPUT	A VU meter of the input level (0 VU = a sine peaking at -18 dBFS).

Doubler

DELAY	The delay time of the copy, 1 - 40 ms, in 1 ms steps (hold Shift for fine adjustment). Short times give width; longer ones start to sound like a separate double.
x2 - x6	LCR and SPREAD only: the right copy's delay is DELAY times this number.
SWAP SIDES	Swaps the left and right copies (in L/R: which side gets the dry signal).

MONO / STEREO	MONO sums the input to mono before doubling; STEREO doubles each side from its own channel.
LEVEL	The level of the delayed copies, -24 to +6 dB.
LOW SPLIT	A high-pass filter (12 dB/oct) on the delayed copies only, 20 - 1000 Hz. Raising it keeps the low end centred and solid while the width happens higher up.
WIDTH	Overall stereo width, 0 - 100%. Lower it to narrow the effect.
DRY	The dry signal on or off. Off gives the delayed copies only, for use on an FX send.
MONO CHK	Collapses the output to mono, so you can check how the doubled sound holds up on a mono system.

Drift

A slow random wander, independent for each copy, like a second take.

DRIFT	One dial, 0 - 100%. It makes each copy's delay time, pan position (wandering inward from its side) and level drift slowly. 50% is the standard amount, 100% twice that, 0 none; the wander also gets a little faster as you turn it up.
--------------	---

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED at the knob's upper right is dark when GUARD is off, red when it is on, and glows brighter while it is catching peaks.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.

SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

Detune

LEFT / RIGHT	Pitch-shifts the left and right copies by -50 to +50 cents. Small opposite settings (for example -8 and +8) give the classic detuned-doubler width. Detune follows the side a copy sits on; in L/R the dry side takes its side's detune too.
---------------------	--

Master

OUTPUT	Output level, -24 to +12 dB.
OUT meters	Output level, left and right, with a floating peak. Click a meter to clear the peak.

The header

PRESETS, arrows and display	PRESETS (or a click on the display) opens the list of factory presets; the ◀ / ▶ keys step through them. The display shows the preset number and name.
Theme picker	Changes the panel's colour theme.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW	Switches between the hardware-style panel and the flat look.

The presets set the mode, delay, multiplier, level, low split, width and the drift settings; DRY, SWAP SIDES, the input mode, MONO CHK, detune, output and the IN / OUT strip are left as they are. The last entry in the list resets everything to the defaults.

Presets

Short	LCR with 6 ms copies, 3 dB down, and a light drift - tight width.
Med	LCR with 12 ms copies at full level and a moderate drift.
Long	LCR with 20 ms copies, 4 dB down, low split at 200 Hz and a little more drift - closer to a separate double.
Classic Haas	L/R with an 18 ms copy and no drift - the textbook Haas widener.
Wide Spread	SPREAD with 12 ms copies, low split at 100 Hz, gentle drift.
Super Wobble	SPREAD with x3 copies and heavy, fast drift of time, pan and level - deliberately seasick.
Vintage ADT	L/R with a 22 ms copy, plenty of time drift and a little level drift at 70% width - tape-style automatic double tracking.
Mono Safe	LCR with x3 copies, 6 dB down, low split at 300 Hz and 70% width, so it stays solid in mono.
Reset all to defaults	Every control back to its starting value.

Tips

- Use MONO CHK while you set DELAY: some delay times cancel badly in mono. Mono Safe is a good starting point when that matters.
- On an FX send, switch DRY off and blend the doubler in with the send level.
- Raise LOW SPLIT on bass-heavy sources so the low end stays in the middle.

- A few cents of opposite DETUNE plus a little DRIFT is often more convincing than a longer delay.

File: gfx-haas-doubler.jsfx